

Cloud LearnX

Mathematics

Circle Theorems — Practice Worksheet

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Topic: Circle Theorems

Instructions:

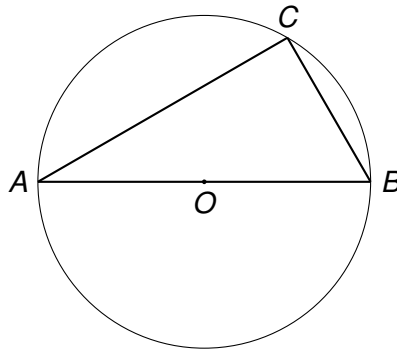
- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** The diagram shows a circle with centre O . AB is a diameter of the circle, and C is a point on the circumference.



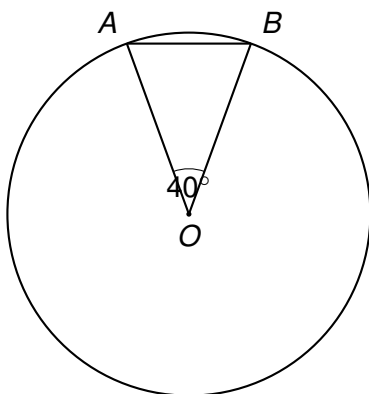
Find the size of angle ACB , giving a reason for your answer.

.....

(2)

(Total for Question 1 is 2 marks)

- 2 O is the centre of the circle below. A and B are points on the circumference, and OA and OB are radii of the circle. A and B are joined to form the triangle OAB .



Angle $AOB = 40^\circ$.

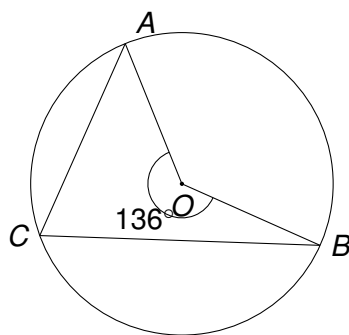
Work out the size of angle OAB . Give a reason for your answer.

$$OAB = \dots\dots\dots^\circ$$

(2)

(Total for Question 2 is 2 marks)

- 3 A , B and C are points on the circumference of a circle with centre O . O is joined to A and to B , forming radii OA and OB .



Angle $AOB = 136^\circ$.

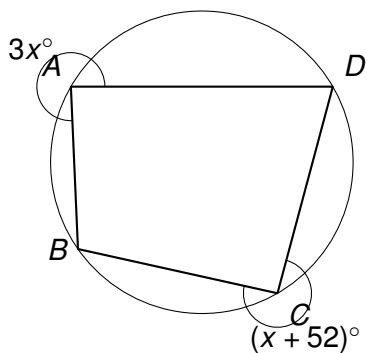
Find the size of angle ACB .

Angle $ACB = \dots\dots\dots^\circ$

(3)

(Total for Question 3 is 3 marks)

4 A , B , C and D are four points on the circumference of a circle, forming the cyclic quadrilateral $ABCD$.



Angle $BAD = 3x$ degrees and angle $BCD = (x + 52)$ degrees.

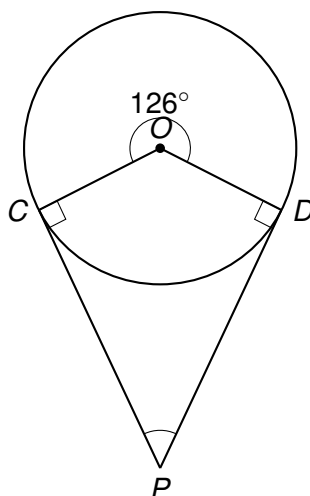
Work out the value of x , and hence find the sizes of angle BAD and angle BCD .

(3)

(Total for Question 4 is 3 marks)

- 5 An engineering student is designing the support frame for a circular satellite dish. The dish's circular support ring has centre O . Two straight support cables are attached to the ring at points C and D , and both cables meet at an external anchor point P , so that PC and PD are tangents to the circle at C and D respectively.

The angle COD at the centre of the ring is 126° .



Work out the size of angle CPD , the angle between the two support cables at P .

$$CPD = \dots\dots\dots^\circ$$

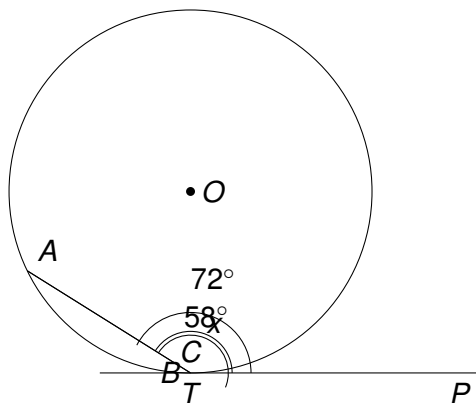
(3)

(Total for Question 5 is 3 marks)

6 A UI/UX designer is studying the optics of a circular lens used in a microscope, so she can render a realistic light-refraction effect in a digital art piece. The cross-section of the lens is modelled as a circle with centre O .

A light ray travels along the straight line PT , which touches the circle at point T (so PT is a tangent to the circle at T). From T , two chords are drawn into the lens: chord TA and chord TC . A third point B lies on the circle, on the opposite side of chord TA from P . Chords TB and AB are also drawn.

The angle between the tangent PT and chord TA is 58° . The angle between the tangent PT and chord TC is 72° .



Find the size of angle ABT , marked x on the diagram.

$x = \dots\dots\dots^\circ$
(4)

(Total for Question 6 is 4 marks)

TOTAL FOR PAPER IS ?? MARKS